

Reflexive

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Parameters

- + $A \in \text{Set}$
- + $R \in A \sim A$

Assumptions

- a. $\text{id}_A \subseteq R$

Observations

1. $(\forall a \mid a \in A : a \mathbf{R} a)$
2. $(\forall a \mid a \in X : (\exists x \mid x \in X : a \mathbf{R} x)) \Leftarrow X \subseteq A$

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Proof of Observation 1

◆ $\text{id}_A \subseteq R$

$\equiv$ Extensionality

$(\forall y, x \mid y \text{id}_A x : y \mathbf{R} x)$

$\equiv$ Definition of identity

$(\forall y, x \mid y = x \wedge x \in A : y \mathbf{R} x)$

$\equiv$ One-point rule

$(\forall x \mid x \in A : x \mathbf{R} x)$

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Proof of Observation 1

◆ $(\forall a \mid a \in X : (\exists x \mid x \in X : a \mathbf{R} x))$

i. $R \in A \sim A$

ii. $(\forall x \mid x \in X : x \in A)$

iii. $(\forall a \mid a \in A : a \mathbf{R} a)$

⊢ **Generalisation, fixing $a \in X$**

◆ $(\exists x \mid x \in X : a \mathbf{R} x)$

iv. $a \in X$

1. $a \in A$

⇐ **∃ introduction**

$(x \in X \wedge a \mathbf{R} x)[x := a]$

≡ **Textual substitution**

$a \in X \wedge a \mathbf{R} a$

≡ **Assumption iv.**

$a \mathbf{R} a$

≡ **Assumption iii. using observation 1.**

true

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